

Analysis Plan — Clustering Analysis on DS-10 (Credit Card Fraud)

Advanced Data Mining, Spring 1404–1405

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Dataset. DS-10 — UCI/Kaggle Credit Card Fraud, 284,807 transactions $\times$ 31 columns (V1–V28 PCA components, `Time`, `Amount`, binary `Class`); 492 fraudulent. ~150 MB.

Central clustering research question. *Do credit-card transactions form density-distinct subpopulations that separate fraudulent from legitimate behaviour, and can those subpopulations be recovered without using the label?*

Expected cluster types. Density-varying / non-convex micro-clusters representing fraud; convex / elliptical macro-clusters representing the bulk of legitimate behaviour; possibly boundary points. The Phase 2 portfolio therefore covers both regimes.

Phase 1 plan. Reproducible ingest (SHA-256, pandera schema) $\rightarrow$ cleaning with decision log (drop exact duplicates) $\rightarrow$ feature engineering (`log1p(Amount)` + cyclic `Time`) $\rightarrow$ 3-strategy scaling comparison with a hybrid `ColumnTransformer` chosen $\rightarrow$ PCA at 95% cumulative variance and UMAP (2D viz, 10D downstream) $\rightarrow$ distance-metric selection (Euclidean headline; Mahalanobis and cosine ablations deferred to Phase 2) $\rightarrow$ Hopkins H and VAT clustering-tendency assessment $\rightarrow$ EDA report.

Phase 2 plan. One algorithm per family: K-Means (partitioning), Ward agglomerative (hierarchical), HDBSCAN (density), GMM with full / diagonal / spherical covariance (model). Determining k via elbow + silhouette + gap statistic + BIC + bootstrap stability. Internal metrics: silhouette, Davies–Bouldin, Calinski–Harabasz, Dunn. External metrics: ARI, NMI, AMI, V-measure, purity (`Class` as ground truth). Stability via 20 seeds + bootstrap; consensus via co-association matrix.

Phase 3 track. *Track 1 — Deep Clustering.* Autoencoder + K-Means baseline $\rightarrow$ DEC $\rightarrow$ custom DEC + InfoNCE contrastive auxiliary (bonus +5). Interpretation via decision-tree rule extraction, SHAP, and counterfactuals (DiCE; bonus +4). Significance via bootstrap CIs + permutation tests (bonus +3). Streamlit dashboard. IEEE LaTeX final report (bonus +2). Non-expert insight brief (bonus +3). Nominal bonus total +17, capped at +15.

Reproducibility. All seeds in `params.yaml`; DVC + Make orchestration; pandera schemas at every stage boundary; deterministic on re-run; `make verify` smoke test in <5 min.